The instability of ghost-scalar wormholes at large dimension

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Abstract

The static, massless, reflection-symmetric wormhole supported by a ghost scalar field (the Ellis–Bronnikov wormhole) has exactly one unstable radial mode, and in D spacetime dimensions its growth rate W has been computed numerically. We derive its large-D expansion by matched asymptotic expansions:

$$W^2a^2 = 2(D-3) - 4\ln 2 + \frac{\pi^2/3 + 4\ln^2 2}{D-3} + O(D^{-2}),$$

with a the throat radius. The first two terms are obtained in closed form. The third is a matching integral evaluated to ten digits and identified by an integer-relation search. The master equation is written in the gauge in which the scalar is unperturbed; two independent integrations of it reproduce the published eigenvalues to nine or more significant figures and confirm the expansion up to $D = 1000$, with a remainder of order D^{-2}. At large D the throat moves rigidly while the mode lives in the nearly flat region outside it, with the profile $\delta r \propto (r/a)\, e^{1-r^2/a^2}$. Because the wormhole is the critical solution of the threshold between collapse to a black hole and inflationary expansion, $1/W$ is also the slope of the logarithmic lifetime law there. Published evolutions in four, five and six dimensions are consistent with it.

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1 Introduction

A traversable wormhole needs matter that violates the null energy condition at its throat [1]. The simplest example is the wormhole of Ellis and Bronnikov [2, 3], held open by a massless scalar field with the wrong sign of kinetic energy (a ghost). It is unstable. González, Guzmán and Sarbach proved that its spherically symmetric perturbations have exactly one exponentially growing mode [4], and evolutions show that, perturbed, the throat either collapses to a black hole or expands into an inflating region [5, 6]. Torii and Shinkai found the same single unstable mode in every dimension they computed, $D = 4$–11, 20, 50 and 100, and noted that the instability becomes faster as D grows [7].

How fast? A fit to their largest-D values gives roughly $W^2a^2 \approx 2D - 8.6$, but no formula has been given. The question is natural for the large-D expansion of general relativity [8], in which the gravitational field of a compact object is confined to a thin layer near its horizon. Here the layer is the region near the throat where the geometry differs from flat space, of width a/D.

This paper derives the large-D expansion of W and the shape of the unstable mode. Section 2 writes the perturbation problem as a single master equation. Section 3 solves it numerically and compares with Ref. [7]. Section 4 carries out the matched expansion, and Sec. 5 describes the mode. Section 6 turns the growth rate into a prediction for the lifetime of near-threshold evolutions and compares it with published runs.

2 The wormhole and its radial perturbations

In $D = n + 2$ dimensions, with $G_{ab} = \kappa T_{ab}$ and a ghost scalar, $T_{ab} = -\left(\partial_a\phi\,\partial_b\phi - \frac{1}{2}g_{ab}(\partial\phi)^2\right)$, the static, massless, reflection-symmetric wormhole is ultrastatic [7]:

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}l^2 + r(l)^2\,\mathrm{d}\Omega_n^2, \qquad r'^2 = 1 - \left(\frac{a}{r}\right)^{2(n-1)}, \qquad \kappa\phi'^2 = \frac{n(n-1)\,a^{2(n-1)}}{r^{2n}}, \tag{1}$$

with $l \in \mathbb{R}$, the throat $r = a$ at $l = 0$, and primes denoting $\mathrm{d}/\mathrm{d}l$. Since $g_{tt} = -1$, t is proper time everywhere.

For spherically symmetric perturbations we write

$$\mathrm{d}s^2 = -e^{2\nu}\mathrm{d}t^2 + e^{2\lambda}\mathrm{d}l^2 + (r + \delta r)^2\mathrm{d}\Omega_n^2, \qquad (\nu, \lambda, \delta r) = (N, \Lambda, Y)(l)\,e^{Wt}, \tag{2}$$

in the gauge $\delta\phi = 0$, which is available everywhere because ϕ' never vanishes. Linearizing the Einstein equations, written for the warped product of the (t, l) plane with the n-sphere, and the scalar equation gives:

- from the tl equation, $\Lambda = Y'/r'$;
- from the scalar equation, $N = \Lambda - nY/r$, after a reparametrization of t removes an integration function of t;
- from the angular equation, the master equation

$$Y'' - \left(\frac{2r''}{r'} + \frac{n\,r'}{r}\right)Y' + \left(\frac{n\,r'^2}{r^2} + \frac{r''}{r}\right)Y = W^2Y. \tag{3}$$

With these, the tt and ll equations hold identically. The throat is a regular singular point of Eq. (3) with exponents 0 and 3; the mode that moves the throat is the even one, normalized by $Y(0) = 1$, and a growing mode ($W > 0$) must decay as $l \to \pm\infty$.

Table 1: W^2a^2 for the growing mode. For $D \le 11$ the values are Torii and Shinkai's [7]; for $D \ge 20$ they are from the r-integration of Sec. 3, with Ref. [7] in parentheses where available. The last columns compare with the three-term expansion (16).

D	W^2a^2	relative difference from (16)	(difference)$\times s^2$
4	1.39705243	2.2	-12
5	2.98495893	0.28	-14
6	4.68662054	5.9×10^{-2}	-10
7	6.46258414	1.05×10^{-2}	-4.3
8	8.28975936	2.4×10^{-3}	2.0
10	12.04426501	6.0×10^{-3}	14
20	31.5751101276 (31.5751101285)	1.3×10^{-3}	48
50	91.3457759034 (91.3457759137)	8.2×10^{-5}	66
100	191.2830176842 (191.2830177297)	9.8×10^{-6}	71
200	391.2543342170	1.2×10^{-6}	72.6
500	991.2379714266	7.5×10^{-8}	73.0
1000	1991.2326553188	8.4×10^{-9}	66 ± 6

Since $\mathrm{d}/\mathrm{d}l = r'\,\mathrm{d}/\mathrm{d}r$, the metric perturbation $\Lambda = Y'/r'$ is simply $\mathrm{d}Y/\mathrm{d}r$. On either side of the throat, with $F(r) = r'^2 = 1 - (a/r)^{2(n-1)}$, Eq. (3) becomes the first-order system

$$\frac{\mathrm{d}Y}{\mathrm{d}r} = \Lambda, \qquad F\,r\,\frac{\mathrm{d}\Lambda}{\mathrm{d}r} = W^2 r Y + \Lambda\left(\frac{rF_r}{2} + nF\right) - \frac{nFY}{r} - \frac{F_r Y}{2}, \tag{4}$$

regular-singular at $r = a$, where $F = 0$. The even mode is the solution analytic in $r - a$. Where $F = 1$ to exponential accuracy, Eq. (4) reduces to $Y'' - (n/r)Y' + (n/r^2)Y = W^2 Y$, whose decaying solution is

$$Y_{\mathrm{far}} = r^{(n+1)/2}\,K_{(n-1)/2}(Wr). \tag{5}$$

3 The numerical spectrum

We solved the eigenvalue problem in two ways. The first integrates Eq. (3) in l from a series start at the throat and shoots on the decay at large l. It is accurate to about 10^{-3} for $D \le 10$, but its start near the throat loses accuracy at large D, where the near-throat layer is thin. The second integrates Eq. (4) in r from a Frobenius series at $r = a$ to $r_m = a\,e^{X/s}$, $s = 2(D-3)$, where $F = 1 - e^{-X}$, and matches Λ/Y there to the logarithmic derivative of Eq. (5). Changing X from 40 to 50, the starting point and the tolerances moves W^2a^2 by 4×10^{-10} at $D = 20$ and 1.5×10^{-6} at $D = 1000$.

Both find exactly one growing mode. The second reproduces Torii and Shinkai's values [7], obtained with a different, gauge-invariant master variable, to nine or more significant figures (Table 1).

4 The large-D expansion

Set $a = 1$, $s = 2(D-3)$, $\varepsilon = 1/s$, and write $W^2 = s + \Omega$ with $\Omega = \Omega_0 + \varepsilon\,\Omega_1 + \dots$. Near the throat we use $R = r^s$, so that $F = 1 - 1/R$ and $r = e^{\varepsilon \ln R}$. Equation (4) becomes, exactly,

$$Y_R = \varepsilon\,\frac{\Lambda r}{R}, \qquad (R-1)\Lambda_R = (1+\varepsilon\Omega)\,rY + \frac{\Lambda}{2R} + \left(\tfrac{1}{2}+\varepsilon\right)\frac{\Lambda(R-1)}{R} - \left(\tfrac{1}{2}+\varepsilon\right)\frac{Y(R-1)}{Rr} - \frac{Y}{2rR}. \tag{6}$$

Expanding $Y = \sum \varepsilon^k Y_k$ and $\Lambda = \sum \varepsilon^k \Lambda_k$ gives at each order

$$(R-1)\Lambda_k' - \tfrac{1}{2}\Lambda_k = h_k(R), \tag{7}$$

where h_k depends on lower orders. The solution analytic at the throat is

$$\Lambda_k = -2h_k(1) + (R-1)^{1/2}\int_1^R \frac{h_k(R') - h_k(1)}{(R'-1)^{3/2}}\,\mathrm{d}R'. \tag{8}$$

At large R it contains $G_k R^{1/2}$, with

$$G_k = \int_1^\infty \frac{h_k(R) - h_k(1)}{(R-1)^{3/2}}\,\mathrm{d}R. \tag{9}$$

But $R^{1/2} = r^{D-3}$ is the far zone's growing solution, the one that Eq. (5) excludes. A growing mode therefore requires $G_k = 0$ at every order, and this fixes Ω.

Orders 0 and 1. At order 0, $h_0 = Y_0/2$ and Y_0 is constant; we normalize $Y_0 = 1$, so $\Lambda_0 = -1$. At order 1,

$$h_1 = \Omega_0 + \ln R - \frac{2(R-1)}{R}, \qquad \Lambda_1 = -2\Omega_0 - 2\ln R, \qquad Y_1 = -\ln R. \tag{10}$$

Here $G_1 = 0$ identically, because $\int_1^\infty \ln R\,(R-1)^{-3/2}\mathrm{d}R = 2\pi = \int_1^\infty 2(R-1)^{-1/2}R^{-1}\mathrm{d}R$.

Order 2. Then $Y_2 = -2\Omega_0 \ln R - \frac{3}{2}\ln^2 R$ and

$$h_2 = \Omega_1 - \frac{2\Omega_0(R-1)}{R} - \Omega_0 \ln R - 2\ln^2 R, \qquad G_2 = -4\pi\Omega_0 - 2J, \tag{11}$$

with, substituting $R = 1 + u^2$ and integrating by parts once,

$$J = \int_1^\infty \frac{\ln^2 R}{(R-1)^{3/2}}\,\mathrm{d}R = 8\int_0^\infty \frac{\ln(1+u^2)}{1+u^2}\,\mathrm{d}u = 8\pi\ln 2. \tag{12}$$

So $G_2 = 0$ gives

$$\Omega_0 = -4\ln 2. \tag{13}$$

Consistently, the solution (8) at this order is

$$\Lambda_2 = -2\Omega_1 + 2\Omega_0\ln R + 4\ln^2 R - 8\Omega_0\,u\arctan u - 16\,u\,K(u), \tag{14}$$

with $u = \sqrt{R-1}$ and $K(u) = \int_0^u \ln(1+t^2)\,(1+t^2)^{-1}\,\mathrm{d}t$, and its $R^{1/2}$ term, $-(4\pi\Omega_0 + 16\pi\ln 2)\,u$, vanishes for the same Ω_0.

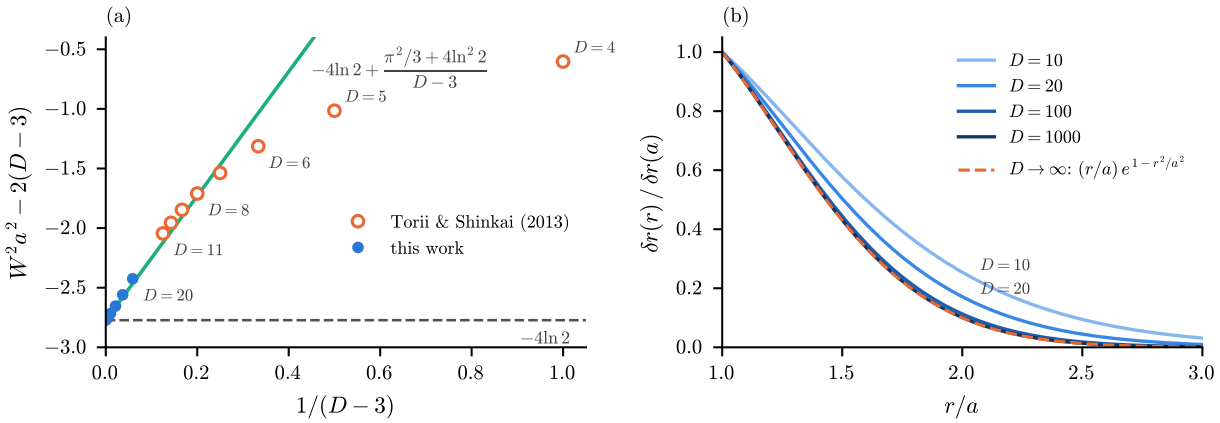


Figure 1: (a) $W^2a^2 - 2(D-3)$ against $1/(D-3)$: Torii and Shinkai's eigenvalues [7] for $D = 4$–11 (open circles) and this work's for $D = 20$–2000 (filled), with the two-term (dashed) and three-term (solid) large-D results (16). (b) The profile $\delta r(r)/\delta r(a)$ of the growing mode for $D = 10$, 20, 100 and 1000, and its large-D limit $(r/a)\,e^{1-r^2/a^2}$ (dashed).

Order 3. Y_3, h_3 and G_3 follow in the same way. Ω_1 enters h_3 only through $-\Omega_1[\ln R + 2(R-1)/R]$, so $G_3 = G_3|_{\Omega_1=0} - 4\pi\Omega_1$. We evaluated G_3 on a grid in $v = \sqrt{\ln R}$, in which every integrand is smooth, with cumulative Simpson integration on three grids. The two finer grids agree to 10^{-9} and give

$$\Omega_1 = 10.4233603788. \tag{15}$$

An integer-relation search (PSLQ [9]) over $\{\Omega_1, \pi^2, \ln^2 2, \ln 2, 1\}$ at tolerance 10^{-9} returns $3\Omega_1 - 2\pi^2 - 24\ln^2 2 = 0$, that is $\Omega_1 = 2\pi^2/3 + 8\ln^2 2 = 10.42336037874$, which differs from the computed value by 6×10^{-11}. We regard this closed form as an identification, not a derivation.

Result. With $s = 2(D-3)$,

$$\boxed{W^2a^2 = 2(D-3) - 4\ln 2 + \frac{\pi^2/3 + 4\ln^2 2}{D-3} + O(D^{-2}).} \tag{16}$$

Table 1 and Fig. 1(a) compare it with the numerical eigenvalues. The difference times s^2 settles near 73 at $D = 200$–500, as an $O(D^{-2})$ remainder should; at $D = 1000$ it is 66 ± 6, within that value's numerical uncertainty. The three terms are within 1.1% of the exact value at $D = 7$–11 and at every larger D computed, and within 10^{-5} (relative) at $D = 100$. At $D = 4$ they are not useful.

Matching. The far solution (5), expanded in the overlap $1 \ll R \ll e^s$, has logarithmic derivative $\Lambda/Y = -1 - \varepsilon(2\Omega_0 + 3\ln R) + O(\varepsilon^2)$, and so does the near-zone solution. The matching at this order holds for any Ω; the condition on Ω comes only from the absence of the growing term.

5 The shape of the mode

The expansion also says where the instability lives. In the layer near the throat, of width $a/(D-3)$ in r, the perturbation is constant to leading order, $Y = Y_0$: the throat moves as a whole, and $\Lambda/Y = \delta r'/\delta r = -1/a$ at the outer edge of the layer. Outside, the geometry is flat up to terms

Table 2: Slope of x_H^-/a against $\ln(1/\Delta E)$ near the threshold. Predicted: $1/(\sqrt{2}\,Wa)$ from Table 1. Measured: Shinkai and Hayward's fitted exponent [5], and two-run slopes from the $\alpha_{\rm GB} = 0$ rows of Shinkai and Torii's Table I [11], with uncertainties from the rounding of that table.

D	$1/(Wa)$	predicted	measured	source	runs
4	0.846	0.598	0.60 (ghost), 0.61 (normal)	[5]	fit
4			0.71 ± 0.14	[11]	$\Delta E/a = 0.03$, 0.10
5	0.579	0.409	0.43 ± 0.02	[11]	$\Delta E/a = 0.15$, 0.61
6	0.462	0.327	0.335 ± 0.011	[11]	$\Delta E/a = 0.38$, 1.50

of order $(a/r)^{2(D-3)}$, and the mode is the decaying solution (5). For $x = Wr$ of order $\nu^{1/2}$, $\nu = (D-3)/2$, the Bessel function behaves as $K_\nu(x) \propto x^{-\nu} e^{-x^2/4\nu}\left[1 + O(\nu^{-1})\right]$, so to leading order

$$\frac{\delta r(r)}{\delta r(a)} = \frac{r}{a}\, e^{1-r^2/a^2}, \tag{17}$$

which has slope $-1/a$ at $r = a$, as the near zone requires. Figure 1(b) shows the numerical profiles approaching this limit.

So at large D the instability is a mode of nearly flat space outside a rigidly moving throat, which acts as a boundary with the fixed condition $\delta r'/\delta r = -1/a$. It extends over a radius of order a, not over the thin layer of width a/D in which the wormhole's geometry is concentrated.

6 Consequence for the collapse–expansion threshold

Shinkai and Hayward evolved the four-dimensional wormhole with Gaussian pulses of either sign of energy and found a sharp boundary between collapse to a black hole and inflationary expansion, which they called "a new kind of critical behaviour reminiscent of that found by Choptuik" [5]. Near it the final black-hole mass tends to a finite value and the time before the outcome grows logarithmically. Both are the signatures of type I critical phenomena [10], whose critical solution is static with exactly one unstable mode. The ghost wormhole is such a solution [4, 7]. The time spent near it is then

$$T \simeq \frac{1}{W} \ln \frac{1}{|p - p_*|} + \text{const}, \tag{18}$$

with p the family parameter and p_* its threshold value; W is the growth rate in proper time at the throat.

Refs. [5, 11] use the null coordinate $x^- = (t - r_*)/\sqrt{2}$, in which a proper-time interval at fixed position is $\Delta x^- = \Delta t/\sqrt{2}$. Their threshold is at zero injected energy, and ΔE is proportional to the distance from it: linear in the amplitude of a ghost pulse, and quadratic in that of a normal pulse, which perturbs the metric only at second order. The predicted slope of the horizon's formation coordinate x_H^- against $\ln(1/\Delta E)$ is therefore $1/(\sqrt{2}\,W)$. Table 2 compares it with the published runs.

The agreement in $D = 4$ is with Shinkai and Hayward's fit over two decades of ΔE; the comparison with the linear growth rate does not appear in Refs. [4, 5]. The higher-dimensional runs of Ref. [11] were not designed to measure the lifetime law, and two runs at moderate amplitude give only a rough slope, but in $D = 5$ and 6 they agree with the prediction within their uncertainties. A bisection to the threshold in $D = 5$ and 6, with the lifetime fitted against $\ln|p - p_*|$, would test Eq. (18) properly. At large D the prediction is $T \simeq a \ln(1/|p - p_*|)/\sqrt{2(D-3)}$.

7 Discussion

Equation (16) answers how the ghost wormhole's instability depends on dimension: its rate grows as $\sqrt{2(D-3)}/a$, with corrections $-4\ln 2$ and $(\pi^2/3+4\ln^2 2)/(D-3)$ to W^2a^2. The leading behaviour has a simple reading. Outside a thin layer the wormhole is flat space, and the mode is the lowest one of flat space with the throat as a boundary. The constant $-4\ln 2$ comes from the layer itself, through the single integral J.

The analysis covers radial perturbations of the massless, reflection-symmetric member of the static family only. The asymmetric members (different asymptotic masses on the two sides) also have a single unstable mode [4], and the same near-zone/far-zone construction should apply to them. Charged wormholes, wormholes supported by a ghost with a potential, and non-spherical perturbations are not considered.

We have not found an analytic expression for this growth rate, or a large-D analysis of these wormholes, in the literature we searched: Refs. [4–7, 11], the recent evolution of Ref. [12], and searches for large-D treatments of wormhole stability. This is not an exhaustive survey.

8 How the results were obtained

The author set the question and scope and reviewed each stage; Anthropic's Claude, run as an agent with a Python environment, did the literature reading, derivations, code and drafting. No claim rests on the assistant's word alone:

- *Master equation.* The reduction of the linearized Einstein–scalar equations to Eq. (3) and the identities for the remaining components were checked symbolically (SymPy), with a control in which a wrong coefficient leaves nonzero residuals.
- *Spectrum.* Two independent integrations, compared with the published table of Ref. [7], which uses a different master variable.
- *Expansion.* Each closed form at orders 1 and 2 was checked by quadrature; Ω_1 was computed on three grids and identified by PSLQ, and the expansion was compared with the numerical eigenvalues up to $D = 1000$.
- *Numbers.* Every number in the tables and text is recomputed by one script from the data files, with checks that fail if a stated agreement does not hold.

These checks are internal; the work has not yet been reviewed by specialists.

Reproducibility. The derivation and integration scripts, the data behind the figure and the number checks are in `notes/wormhole_largeD/` of the accompanying repository, https://github.com/nmoinvaz/negative-mass, and the figure is made by `paper/figures/make_figures.py`.

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